

21. Swap the Cases of a Character Column

Aim

To write a Pandas program to swap the cases of characters in a specified column of a DataFrame.

Dataset

Name	City
Anu	Chennai
Bala	MADURAI
Cathy	Coimbatore
David	CHENNAI
Esha	Bangalore

Python code:

```
1 import pandas as pd
2 data = {
3     "Name": ["Anu", "Bala", "Cathy", "David", "Esha"],
4     "City": ["Chennai", "MADURAI", "Coimbatore", "CHENNAI", "Bangalore"]
5 }
6 df = pd.DataFrame(data)
7 print("Original DataFrame:")
8 print(df)
9 # Swap the case of City column
10 df["City"] = df["City"].str.swapcase()
11 print("\nAfter Swapping Case:")
12 print(df)
```

Output

	Name	City
0	Anu	CHENNAI
1	Bala	madurai
2	Cathy	COIMBATORE
3	David	chennai
4	Esha	bANGALORE

Result:

The cases of characters in the specified column were successfully swapped.

22. Draw a Line with Labels and Title

Aim

To write a Python program to draw a line chart with suitable labels for the X-axis, Y-axis, and title.

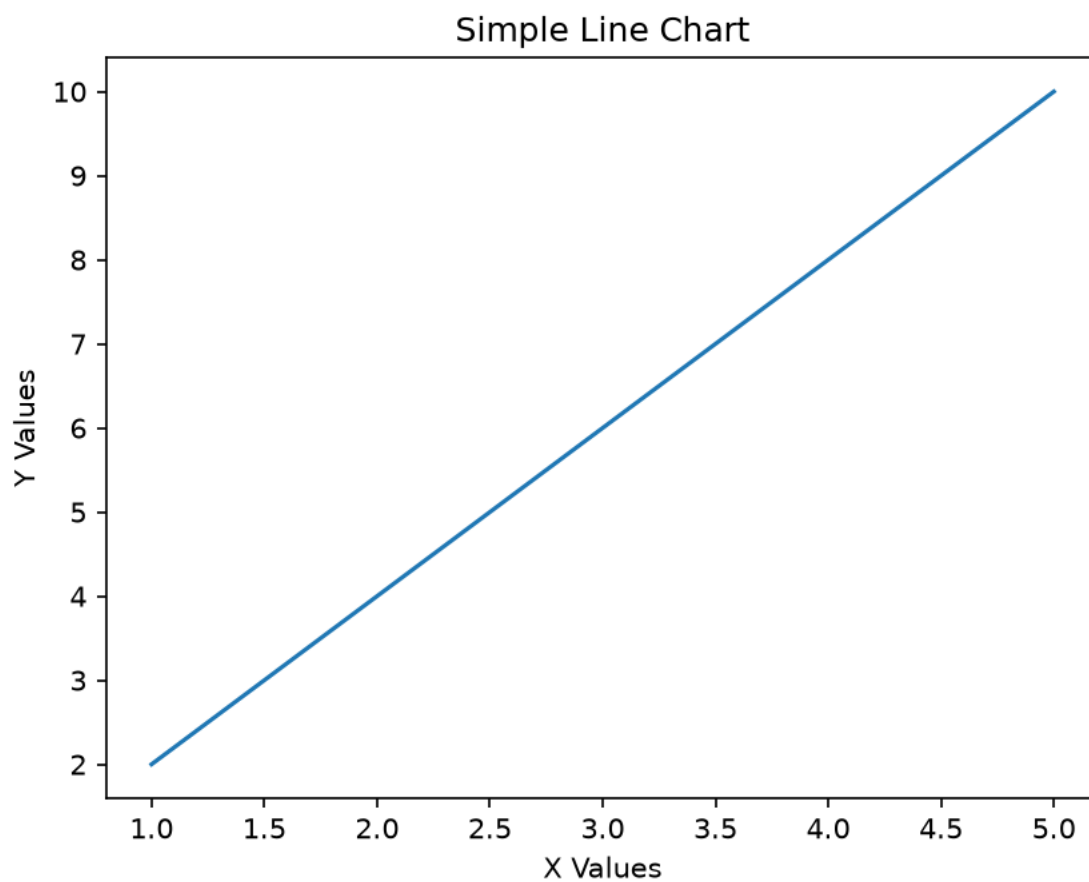
Dataset

X	Y
1	2
2	4
3	6
4	8
5	10

Python code:

```
❖ 1 import matplotlib.pyplot as plt
2 x = [1, 2, 3, 4, 5]
3 y = [2, 4, 6, 8, 10]
4 plt.plot(x, y)
5 plt.xlabel("X Values")
6 plt.ylabel("Y Values")
7 plt.title("Simple Line Chart")
8 plt.show()
```

Output:



Result:

A line chart was successfully created with appropriate axis labels and title.